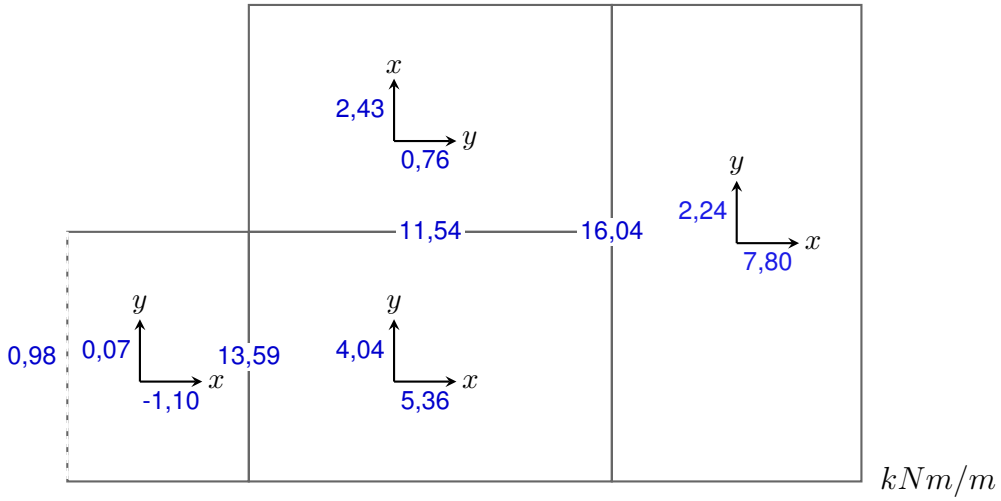
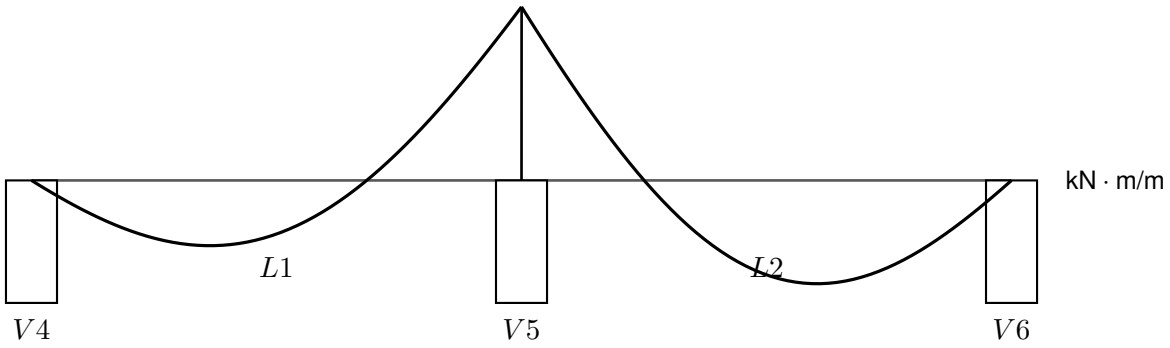
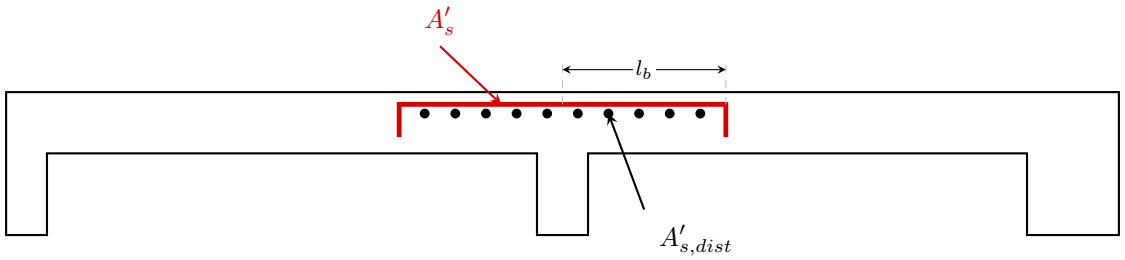
Aula 04b — 10 de setembro de 2026

9. Dimensionamento armaduras negativas



L1-L2 L2-L4





a) Momento de projeto

$M'k = 16,04\ kNm/m$

$\gamma_c = 1,4$ (ABNT NBR 6118, 2023, tabela 12.1)

$M'_d = \gamma_c\ M'k = 1,4 \times 16,04 = 2245,6\ kNcm/m$

b) Altura útil

$$d' = h - c - \frac{\phi_\ell}{2} = 10 \cdot 10^{-2} - 20 \cdot 10^{-3} - \frac{10 \cdot 10^{-3}}{2} = 7,5\ cm$$

c) Taxa mínima de armadura

$\rho_s \geq \rho_{min}$(ABNT NBR 6118, 2023, tabela 17.3 e tabela 19.1)

$$\frac{A_s}{b_w\ h} \geq \frac{0,179}{100} \longleftrightarrow A_s min = \frac{0,179 \times 100 \times 10}{100} = 1,79\ cm^2/m$$

d) Área de aço

$$K_c = \frac{b\ d'^2}{M_d} = \frac{100 \times 7,5^2}{2245,6} = 2,50\ cm^2/kN$$

$\beta_x = 0,24 < 0,45 \rightarrow$ armadura simples, domínio 2 (SILVA et al., 2023, tabela 1.1a)

$$K_s = \frac{A_s\ d'}{M_d} \longleftrightarrow A_s = \frac{0,025 \times 2245,6}{7,5} = 7,49\ cm^2/m$$

e) Armadura de distribuição negativa (ABNT NBR 6118, 2023, item 20.1)

A armadura secundária ou armadura de distribuição serve, apenas, como elementos construtivo.

$$A'_{s,dist} \geq \begin{cases} 20\% A_s = \dfrac{20}{100}\ 7,49 = 1,50\ cm^2 \\ 0,9\ cm^2/m \end{cases}$$

f) Determinação da bitola e do espaçamento

Sempre considerar a altura da menor laje.

$$\phi_{\ell,max} = \frac{h}{8} = 1,25\ cm = 12,5\ mm$$

$$S_p \leq \begin{cases} 2h = 2 \times 10 = 20\ cm \\ 20\ cm \end{cases} \qquad S_d \leq 33\ cm$$

$$A_\phi = \pi\left(\frac{D}{2}\right)^2 \qquad A_\phi = \pi\left(\frac{D}{2}\right)^2$$

$$A_\phi = \pi\left(\frac{12,5 \cdot 10^{-3}}{2}\right)^2 = 1,23\ cm^2 \qquad A_\phi = \pi\left(\frac{6,3 \cdot 10^{-3}}{2}\right)^2 = 0,31\ cm^2$$

$$S_{\phi,p} = \frac{A_\phi}{A_s}\ 100 \qquad S_{\phi,d} = \frac{A_\phi}{A_s}\ 100$$

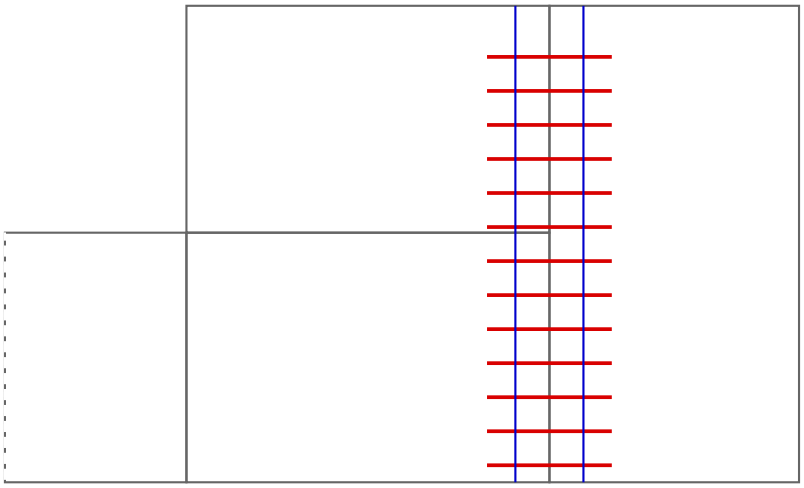
$$S_{\phi,d} = \frac{1,23}{7,49}\ 100 = 16,42 \approx 16\ cm \qquad S_{\phi,d} = \frac{0,31}{1.79}\ 100 = 17,32 \approx 17\ cm$$

$\phi_\ell = 12,5mm$ c/ $16\ cm$ $\qquad$ $\phi_\ell = 6,3mm$ c/ $17\ cm$

ℓ_x vai ser o maior vão efetivo das lajes contínuas.

$$\ell_x \geq \begin{cases} \ell_{L1,L2} \geq \begin{cases} \ell_{x,L1} = 256\ cm \\ \ell_{x,L2} = 446\ cm \end{cases} \\ \ell_{L2,L4} \geq \begin{cases} \ell_{x,L2} = 446\ cm \\ \ell_{x,L4} = 476\ cm \end{cases} \end{cases}$$

$$a_\ell = 0,25\ \ell_x = 0,25 \times 476 = 119\ cm$$



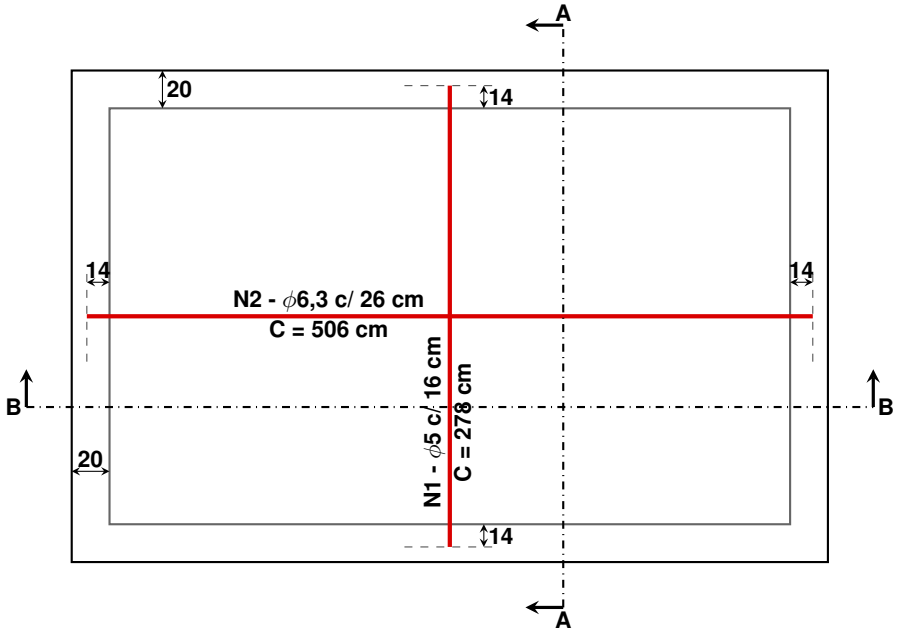
kNm/m

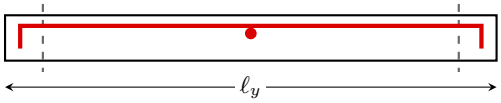
L1-L4

L3-L4

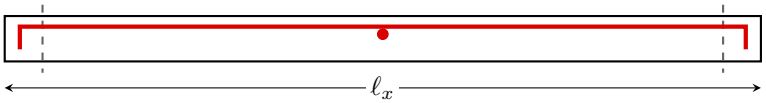
10. Detalhamento das armaduras

L1





CORTE AA



CORTE BB

a) Comprimento de ancoragem básico (ABNT NBR 6118, 2023, item 9.4.2.4)

i. Resistência da aderência de cálculo (ABNT NBR 6118, 2023, item 9.3.2)

$f_{ctk,inf} = 0,7\ f_{ct,m}$ (ABNT NBR 6118, 2023, item 8.2.5)

$f_{ct,m} = 0,3 f_{ck}^{2/3}$ (ABNT NBR 6118, 2023, item 8.2.5)

$\gamma_c = 1,4$ (ABNT NBR 6118, 2023, tabela 12.1)

$$f_{ctd} = \frac{f_{ctk,inf}}{\gamma_c} \text{(ABNT NBR 6118, 2023, tabela 8.2)}$$

$$f_{ctd} = \frac{0,7 \times 0,3 \times 25^{2/3}}{1,4} = 1,28\ MPa$$

$\eta_1 = 2,25$ (ABNT NBR 6118, 2023, tabela 8.2)

$\eta_2 = 1$ (ABNT NBR 6118, 2023, item 9.3.1)

$\eta_3 = 1$ (ABNT NBR 6118, 2023, item 9.3.2.1)

$$f_{bd} = \eta_1\ \eta_2\ \eta_3 f_{ctd} = 2,25 \times 1 \times 1 \times 1,28 = 2,89\ MPa$$

$\gamma_s = 1,15$ (ABNT NBR 6118, 2023, tabela 12.1)

$$f_{yd} = \frac{f_{yk}}{\gamma_s}$$

$$\ell_b = \frac{\phi}{4}\frac{f_{yd}}{f_{bd}} \geq 25\phi$$

$$\ell_b = \frac{\times 0,5}{4}\frac{\frac{500}{1,15}}{2,89} \geq 25 \times 0,5$$

$$\ell_b = 18,81 \geq 12,5$$

b) Comprimento de ancoragem necessário (ABNT NBR 6118, 2023, item 9.4.2.5)

1 para ficar a favor da segurança.

$\alpha = 1$ (ABNT NBR 6118, 2023, item 9.4.2.5)

$$\ell_{b,nec} = \alpha\ \ell_b \frac{A_{s,calc}}{A_{s,ef}} \geq \ell_{b,min}$$

$$\ell_{b,nec} = \alpha\ \ell_b \frac{A_{s,calc}}{A_{s,ef}} \geq \begin{cases} 1 \\ 0,3\ell_b = 0,3 \times 18,81 = 5,64\ cm \\ 10\phi = 10 \times 0,5 = 5\ cm \\ 10\ cm \end{cases}$$

$$\ell_{b,nec} = 1 \times 18,81 = 18,81 \geq 10\ cm$$

L2

L3

L4

Referências

Associação Brasileira de Normas Técnicas. *NBR 6118: Projeto de estruturas de concreto*. Rio de Janeiro, 2023. 242 p.

SILVA, G. d. S. et al. *Tabelas Kc e Ks para fck < 90 MPa*. 1. ed. Barra do Garças: Universidade Federal de Mato Grosso, 2023.